

BREAK INTO AEROSPACE

From Automotive & Diesel Mechanics to Aerospace

Career Transition Guide

Targeted for the 2025–2026 Recruitment Cycle

What This Document Covers: This guide is written for people who have built their technical skills through automotive and diesel mechanical work — apprentice-trained mechanics, City & Guilds or NVQ-qualified technicians, IMI-certified professionals, ASE-certified mechanics, heavy diesel technicians, plant and agricultural machinery engineers, and anyone who has spent years diagnosing, repairing, and overhauling mechanical systems on vehicles and machines. It is specifically not written for mechanical engineers with university degrees: that guide exists separately. This guide is for the person who has built their competence through doing — through pulling engines, diagnosing faults with a multimeter and a scan tool, replacing injectors, setting valve timing, and understanding why a turbocharged diesel that was running correctly yesterday isn't running correctly today.

The honest starting point: automotive and diesel mechanics are better positioned for aerospace maintenance and technical roles than almost anyone in the general population — and they almost never know it. The skills that the automotive mechanic uses daily — systematic fault isolation, reading and following technical documentation, mechanical assembly to dimensional tolerances, understanding how systems interact, recognising when a measurement is wrong and why — are precisely the skills that aircraft maintenance organisations look for in their technicians. The aircraft is not a car. But the aircraft mechanic's job, at its most fundamental level, is the same job: understand the system, identify the fault, fix it correctly, verify the fix, and document what you did. Every mechanical technician who has done this honestly for five years in automotive or diesel is doing the same thing that an aircraft maintenance engineer does, in a different physical context with a different regulatory framework.

The regulatory framework is the primary gap — and it is a specific, learnable gap. The Part-66 licence (for European aircraft maintenance) or the FAA A&P certificate (for US aircraft maintenance) is the professional credential that authorises an individual to certify that aircraft maintenance has been correctly performed. These credentials have specific examination requirements, specific practical experience requirements, and specific procedural discipline requirements that the automotive mechanic must specifically acquire. But the examination content — the physics, the mechanical principles, the electrical fundamentals, the engine theory — overlaps substantially with what an automotive or diesel technician already knows. The compression ignition cycle that a diesel mechanic understands in detail is the same cycle that a Part-66 examiner tests in the aircraft piston engine module. The turbine engine theory that Part-66 tests in Module 15 (Gas Turbine Engine) describes a thermodynamic cycle — the Brayton cycle — that the diesel mechanic can learn from a position of understanding thermodynamics already.

This guide maps the translation specifically: what the automotive and diesel mechanic already knows, what the Part-66 or A&P pathway requires that is genuinely new, and how to prepare for the specific interview and examination demands of the transition. It is written with equal respect for the vocational background and for the regulatory framework of aerospace — because both are demanding, both produce competence, and both deserve to be understood clearly before the decision to transition is made.

The aerospace roles most directly accessible from an automotive and diesel mechanics background: licensed aircraft maintenance engineer Cat B1.1 (the airframe and mechanical systems licence category — the most directly relevant to mechanics with piston and turbine engine experience), aircraft maintenance engineer Cat A (the base maintenance support category — the entry point that does not require the full B1 licence), aircraft engine fitter or powerplant technician (supporting licensed engineers in engine change, engine run, and powerplant maintenance), ground support equipment engineer (maintaining the tractors, tugs, GPUs, belt loaders, and other mechanical equipment that services aircraft on the apron — the most immediately accessible role for automotive and diesel mechanics), aircraft component mechanical overhaul (overhauling mechanical components — actuators, valves, gear boxes, pumps — in Part-145 component maintenance organisations), heavy maintenance mechanical technician (working on structural and mechanical systems in base maintenance checks), and airframe sheet metal and mechanical fitter (fabrication, installation, and repair of airframe structure and mechanical systems in both line and base maintenance).

The employer landscape: airline engineering departments (British Airways Engineering, easyJet Engineering, Ryanair Engineering, Lufthansa Technik, Air France Industries, Iberia Maintenance), independent MRO organisations (SR Technics, Haeco, ST Engineering, AAR Corp, StandardAero), ground handling agents with technical service capability (Swissport Technical Services, Menzies Aviation technical), ground support equipment maintenance contractors (TLD, Charlotte, JBT Aerotech — the GSE manufacturers' service networks, plus airline ground operations engineering), component overhaul facilities (Eaton Aerospace component MRO, Parker Aerospace component MRO, Safran Landing Systems overhaul), and apprenticeship and direct-entry training programmes at airlines that specifically recruit from trade backgrounds (British Airways Engineering

PART I: WHAT AEROSPACE EMPLOYERS ACTUALLY SEE IN AN AUTOMOTIVE AND DIESEL MECHANIC

The Genuine Advantages — What Transfers Directly

The automotive and diesel mechanic's most directly transferable advantage — and the one most consistently undervalued by both parties — is the diagnostic mindset. The experienced mechanic who takes a vehicle with an intermittent starting fault and systematically tests the fuel pressure, the cranking speed, the injector timing, the coolant temperature sensor output, and the compression in each cylinder before forming a conclusion is doing exactly the same systematic elimination process that aircraft fault isolation requires. The aircraft maintenance manual's fault isolation logic tree is the same structured diagnostic process that a competent mechanic applies intuitively — the AMM just makes it explicit and mandatory.

Engine knowledge transfers substantially. The automotive mechanic who understands how a turbocharged diesel engine works — the intake manifold pressure, the exhaust gas temperature as an indicator of combustion completeness, the relationship between boost pressure and fuel quantity for smoke-free combustion, the turbocharger's response to sudden load changes, and the consequences of oil contamination in the intercooler — has developed the engineering intuition for gas path analysis that is directly applicable to gas turbine engine maintenance. Aircraft piston engines (used in training aircraft, light aircraft, and some general aviation) use the same four-stroke Otto cycle as a petrol car engine, with the same valve timing, ignition timing, and mixture control principles. Aircraft turbine engines add the complexity of compressor, combustion, and turbine stage analysis — but the thermodynamic intuition for how gas behaves under compression and expansion, which diesel mechanics develop from working with turbocharged systems, is a genuine head start.

Hydraulic and pneumatic knowledge transfers directly. The automotive mechanic who has worked on power steering systems, ABS hydraulic modulators, clutch hydraulics, and air brake systems understands fluid power at a practical level. Aircraft hydraulic systems are more complex and more safety-critical than automotive hydraulic systems, but the basic principles — pressure, flow, actuator force and velocity, the function of valves and accumulators — are the same. The diesel technician who has diagnosed a hydraulic braking system fault using a pressure gauge and systematic valve isolation has the diagnostic approach for aircraft hydraulic troubleshooting.

Documentation discipline, when it exists in automotive, transfers directly — and where it does not exist, it is the most important thing to develop specifically. The automotive workshop that uses a proper job card system — recording what was found, what was done, what parts were used, and what was verified — is doing the same documentation activity that aircraft maintenance requires, but with different forms and much higher stakes. The aircraft technical log is the aircraft's legal health record, and every entry in it is a permanent regulatory document. The mechanic who has been meticulous about job card documentation in automotive is already practising the documentation discipline that aviation requires — the additional step is understanding why the documentation has regulatory rather than merely commercial significance.

The Specific Gaps — What Does Not Transfer Without Work

The Part-66 licence or FAA A&P certificate — the regulatory credential that separates the aircraft certifying engineer from everyone else. Without a Part-66 licence, no one can sign the Certificate of Release to Service that legally authorises an aircraft to fly after maintenance. Without an FAA A&P certificate, no one can legally approve the return to service of a US-registered aircraft after maintenance. These credentials are the first and most important specific gap for every automotive and diesel mechanic entering aircraft maintenance. Part-66 Category B1.1 (for turbine aircraft mechanical systems and airframe) requires passing 11 module examinations at a minimum score of 75%, plus 3 years of supervised practical aircraft maintenance experience. The FAA A&P requires passing three written knowledge tests and practical oral and performance examinations, plus 30 months of documented practical experience. Understanding the pathway, the timeline, and the financial commitment of these credentials before beginning the transition is essential.

Aircraft-specific engine knowledge — the gas turbine and the certified aircraft piston engine. The automotive mechanic knows petrol and diesel engines. Aircraft engines are different in two specific ways: certified aircraft piston engines (Lycoming, Continental, Rotax) are built and maintained to different specifications than automotive engines — they use specific fuel grades (Avgas 100LL rather than petrol), specific

oil grades (ashless dispersant oils, not the API-rated automotive oils), and are maintained to specific AMM procedures that differ from automotive practice in their torque specifications, their mandatory replacement items, and their inspection requirements. Gas turbine engines — the compressor-combustor-turbine powerplants that power all commercial and most military aircraft — are genuinely different machines from reciprocating engines, operating on the Brayton cycle rather than the Otto or Diesel cycle. The diesel mechanic who understands thermodynamics well has an advantage in learning gas turbine theory; but the physical components (compressor stages, combustion liners, turbine blades, hot section inspection methodology) are new and require specific study.

The Aircraft Maintenance Manual and the aviation technical documentation system. Aircraft maintenance does not use manufacturer's workshop manuals of the type familiar in automotive. Every maintenance task on a certificated aircraft must be performed in accordance with the approved maintenance data — the Aircraft Maintenance Manual (AMM), the Illustrated Parts Catalogue (IPC), the Structural Repair Manual (SRM), the Component Maintenance Manual (CMM), and the Wiring Diagram Manual (WDM). These documents use the ATA chapter numbering system (ATA 71 for Power Plant, ATA 32 for Landing Gear, ATA 27 for Flight Controls), a specific task format with numbered steps, and specific tooling and consumable references that must be exactly followed. The habit of improvising or using a general-purpose approach that many automotive mechanics have developed — applying professional judgement to deviate from the specification when it seems reasonable — is specifically not appropriate in aircraft maintenance, where every deviation from the approved data is a regulatory event requiring engineering authority.

The Part-145 quality culture — documentation before action, reporting rather than concealing. The Part-145 approved maintenance organisation operates under a quality management system that imposes specific procedural requirements with no automotive equivalent. Every maintenance action must be traceable to an approved data reference. Every tool used must be from the organisation's calibrated tool inventory, signed out and back in. Every discrepancy discovered — a crack found during an inspection that was not the purpose of the visit, a fastener found with incorrect torque, a component found with an incorrect part number — must be documented and reported through the organisation's non-conformance system rather than quietly corrected. The automotive mechanic who has worked in an environment where the unspoken rule is "fix it and don't tell the customer more than you have to" must specifically develop the opposite reflex for aerospace — the obligation to document everything, report everything, and never conceal a finding regardless of the inconvenience it creates.

Aircraft-specific materials, fasteners, and hardware. Aircraft are assembled from specific approved materials (aerospace-grade aluminium alloys, titanium fasteners, Inconel bolts in hot sections) using specific approved hardware (NAS and MS standard fasteners in specific materials and finishes) installed with specific procedures (specified torque values, specific safety wiring patterns, specific thread lubricant or thread-locking requirements). The automotive mechanic who understands the general principle of tightening a bolt to the right torque must specifically learn that aircraft fastener torque specifications differ from automotive practice (aircraft bolt torques are often given in inch-pounds rather than Newton-metres, and the torque value for a given bolt diameter depends on the fastener material, the installation location, and whether a lubricant or thread-locker is specified). Improper torque — too low or too high — is a common aircraft maintenance error that specifically contributes to fastener loosening, structural fatigue cracking at fastener holes, and fastener failure in service.

Composite structure — CFRP, GFRP, and the specific repair and inspection requirements. Modern aircraft use composite materials (carbon fibre reinforced polymer and glass fibre reinforced polymer) for structural components ranging from control surfaces to fuselage panels to wing structures. Composites behave differently from metals in almost every respect relevant to maintenance: they do not dent in a visible way when impacted (impact damage may produce internal delamination that is invisible on the surface), they cannot be inspected by the same NDT methods as metallic structure in all cases, and their repair requires specific approved repair materials and specific cure cycle procedures that differ fundamentally from metal repair. The automotive mechanic who has worked on fibreglass body panels has some practical familiarity with composite materials, but the aerospace composite repair framework — the Structural Repair Manual allowable damage limits, the approved repair materials and procedures, and the NDT inspection required after repair — is specifically new.

By Career Stage

Apprentice-Trained or Early Career Mechanic (0–5 years)

The recently qualified mechanic has strong foundational skills and the most flexibility. The direct entry route to aerospace is either the aviation apprenticeship (some airlines and MROs run specific apprenticeship programmes that take automotive and engineering technician entrants and train them to Part-66 Cat A or B1 standard over 3–4 years), the GSE engineer role (which allows aviation sector experience to be accumulated while Part-66 study is pursued), or a direct entry Part-147 training course that converts the existing mechanical background into Part-66 examination credits through a structured programme.

Experienced Mechanic (5–15 years)

The experienced mechanic has the diagnostic confidence, the tool competence, and the mechanical system intuition that aerospace values most highly. The translation task is entirely regulatory and procedural: understanding the Part-66 pathway, studying the aviation-specific examination content (gas turbine theory, aircraft systems, aviation legislation), and developing the aircraft-specific documentation and procedural discipline. Experienced mechanics typically find the Part-66 examination content less daunting than they expect — the mechanical and thermodynamic foundations overlap significantly with their existing knowledge.

Senior Mechanic / Workshop Supervisor

The senior mechanic who has led teams, managed workshop quality, and developed diagnostic procedures has supervisory and leadership competencies that map directly to shift leader and lead technician roles in MRO operations. The regulatory vocabulary gap is the primary obstacle — senior mechanics who specifically address the Part-66 pathway and the Part-145 quality framework are strongly competitive for experienced technician positions.

Qualification Mapping — Where Your Background Fits Best

Light Vehicle (petrol and diesel) Mechanic → Aircraft Piston Engine Technician, Light Aircraft Maintenance (GA), Part-66 Cat B1.2 (Piston Aircraft)

Piston engine overhaul, ignition systems, fuel systems, electrical troubleshooting — transfer with aviation-specific calibration. The aerospace addition: certified aircraft piston engine specifications (Lycoming, Continental), Avgas 100LL fuel requirements, aircraft ignition system specifics (magneto-based, not distributor), and the specific maintenance intervals and airworthiness directive compliance for certificated aircraft.

Diesel and Heavy Vehicle Mechanic → Part-66 Cat B1.1 (Turbine Aircraft Mechanical), Aircraft Engine Change, Powerplant Technician

Turbocharged engine systems, fuel injection, air system management, heavy component removal and installation — transfer with turbine-specific extension. The aerospace addition: gas turbine engine theory (Brayton cycle, compressor and turbine stage analysis, EGT monitoring), specific engine change procedures (dressing, installation, alignment, and borescope inspection), and the Part-145 engine run procedures.

Plant and Agricultural Machinery Engineer (excavators, agricultural tractors, mining equipment) → Ground Support Equipment Engineer, Aircraft Tug Maintenance, Airport Equipment

Heavy hydraulic systems, diesel engine maintenance, complex multi-system diagnosis — transfer very directly. GSE roles require the least aviation-specific technical addition of any aerospace maintenance role, while providing real aviation sector experience. The aerospace addition: IATA AHM GSE standards, aircraft proximity safety procedures, airside safety environment, and the specific electrical and hydraulic systems of aircraft-servicing equipment.

Racing or Performance Engine Builder → Aircraft Piston Engine Overhaul, Engine Test Cell Technician

Precision engine assembly, torque sequences, dimensional measurement of engine components, performance analysis from engine data — transfer directly. The aerospace addition: FAA or EASA approved overhaul data references (the Engine Maintenance Manual and the Overhaul Manual), airworthiness release documentation, and the specific inspection criteria for certificated aircraft engine components.

Marine Diesel Engineer (ships and boats) → Aircraft Gas Turbine Technician, Turboshaft Engine Maintenance (Rotorcraft)

Large diesel engine systems, turbocharger maintenance, fuel and lubricating oil systems, multi-engine management — transfer with significant overlap to aircraft turbine systems. The aerospace addition: gas turbine theory and operation (the Brayton cycle versus the Diesel cycle), jet fuel versus marine diesel fuel properties, and the aircraft-specific ATA documentation system.

Vehicle Electrician / Auto Electrician → Cat B2 (Avionics) Pathway, Aircraft Wiring, GSE Electrical

Automotive electrical fault diagnosis, CAN bus understanding, ECU diagnostics, wire repair and connector installation — transfer with ARINC bus and aircraft electrical architecture addition. The aerospace addition: 115V 400Hz AC aircraft power system architecture, ARINC 429 and 664 bus protocols, DO-160 environmental requirements for airborne electrical equipment, and Part-66 Module 3 and 4 content for the Cat B2 licence pathway.

PART II: 20 SPECIFIC PREP TASKS FOR THE TRANSITION**FOUNDATIONAL — DO THESE FIRST****1. Understand the Part-66 licence pathway — specifically Cat B1.1 for turbine aircraft, and how your automotive background maps to the modules**

3–4 hours

Skill: The primary regulatory credential for aircraft maintenance — understanding the pathway before committing to it

Why the field cares: Part-66 Category B1.1 authorises the holder to certify maintenance on turbine-powered fixed-wing aircraft — the aircraft flown by all commercial airlines. It is the natural target for the diesel and heavy mechanic because the mechanical systems knowledge, the turbine engine thermodynamics, and the hydraulic and pneumatic systems overlap substantially with automotive and diesel heavy vehicle competence. The B1.1 examination requires passing 11 modules: M1 (Mathematics — basic algebra, trigonometry), M2 (Physics — mechanics, thermodynamics, electricity), M3 (Electrical Fundamentals), M4 (Electronic Fundamentals), M5 (Digital Techniques — typically limited depth for B1), M6 (Materials and Hardware — metals, composites, fasteners, standard practices), M7A (Maintenance Practices — turbine aircraft), M8 (Basic Aerodynamics), M9A (Human Factors), M10 (Aviation Legislation), and M11A (Turbine Aeroplane Aerodynamics, Structures and Systems). For the diesel mechanic: M2 Physics overlaps substantially with engine thermodynamics knowledge; M6 Materials and Hardware overlaps with workshop practice; M7A Maintenance Practices overlaps with general maintenance discipline; M11A Gas Turbine Systems covers the aircraft engine in detail and requires the most new study. M1 Mathematics, M8 Aerodynamics, M9A Human Factors, and M10 Aviation Legislation are largely new content. The 3-year experience requirement under Part-66 supervision means that the examination programme should be pursued in parallel with an aviation role — typically a Cat A position or a GSE engineer position — that provides the supervised aircraft maintenance exposure while the B1 examinations are being completed.

Done signal: You can state which Part-66 modules overlap substantially with your automotive or diesel background (M2 Physics for thermodynamics and mechanics, M6 for materials and hardware, M7A for maintenance practices) and which require the most new study (M10 Aviation Legislation, M11A Gas Turbine Systems, M8 Aerodynamics). You have identified an EASA Part-147 approved training organisation that offers B1.1 examination preparation in your area, know the cost and duration of their programme, and have a realistic timeline for completing the examination programme alongside employment. You understand that the B1.1 basic licence requires 3 years of supervised practical experience and that a type rating on a specific aircraft type is additionally required before CRS sign-off authority is granted.

2. Understand the gas turbine engine — the Brayton cycle, the main engine sections, and how it differs from a diesel

6–8 hours

Skill: The foundational engine knowledge that is genuinely new for most automotive and diesel mechanics

Why the field cares: The gas turbine engine is the most important specific knowledge gap for the automotive or diesel mechanic entering aircraft maintenance — because the turbine engine is what makes commercial aircraft different from anything in automotive or diesel experience. The diesel mechanic knows the compression ignition cycle: air is compressed in the cylinder, fuel is injected,

6–8 hours

the heat of compression ignites the fuel, the expanding combustion gases push the piston down, and the piston's mechanical work drives the crankshaft. The gas turbine operates completely differently: air enters the engine continuously (not in pulses), is compressed by a series of rotating compressor stages (not a piston), is mixed with continuously injected fuel and burned in a combustion chamber (not a cylinder), and the hot gases expand through a turbine (which extracts work to drive the compressor) before being exhausted at high velocity (producing thrust). The key difference from the diesel mechanic's perspective: the gas turbine has no pistons, no valves, no crankshaft, and no ignition timing — but it does have compressor and turbine blades that can erode and crack, combustion liners that can crack and burn through, fuel nozzles that can coke and block, and turbine discs that are life-limited by the number of engine cycles they have accumulated. Learning the gas turbine's components, failure modes, and maintenance practices is the most important specific study task for the automotive mechanic targeting commercial aircraft maintenance.

Done signal: You can describe the Brayton cycle — the four thermodynamic processes (isentropic compression, constant pressure heat addition, isentropic expansion, constant pressure heat rejection) — and explain how each process is implemented in the physical components of a turbofan engine (the fan and compressor sections for compression, the combustion section for heat addition, the turbine section for expansion, and the exhaust nozzle for heat rejection). You can explain what bypass ratio is and why high-bypass turbofans are more fuel-efficient than low-bypass jets (the fan accelerates a large mass of air by a small velocity increment, which is thermodynamically more efficient than accelerating a small mass by a large velocity increment). You can name the four main sections of a turbofan engine (fan, compressor, combustor, turbine), identify the hottest section (the high-pressure turbine inlet, where temperatures exceed the melting point of the turbine disc material — requiring cooling air from the compressor), and describe what borescope inspection is used to examine (the internal surfaces of the combustor and turbine, accessed through borescope ports without engine disassembly).

4–5 hours

3. Learn the ATA chapter system and how the AMM is structured — the primary reference for all aircraft maintenance

Skill: The documentation system that replaces the workshop manual in aircraft maintenance

Why the field cares: Every maintenance task on a certificated aircraft must be performed in accordance with the Aircraft Maintenance Manual, and the AMM is organised by ATA chapter numbers. ATA 71 is Power Plant. ATA 32 is Landing Gear. ATA 27 is Flight Controls. ATA 24 is Electrical Power. ATA 28 is Fuel. These chapter numbers are the same on every commercial aircraft. The mechanic who knows that a fuel system fault requires ATA 28 reference and a landing gear fault requires ATA 32 reference can navigate the AMM efficiently from the first day on the job. Within each ATA chapter, the AMM provides: a Description and Operation section (explaining how the system works), a Fault Isolation section (the troubleshooting procedure for reported faults), a Maintenance Practices section (standard procedures applicable across multiple tasks), and specific Removal and Installation tasks for each component. The habit of always finding and reading the AMM task before beginning any aircraft maintenance work — and following it step-by-step rather than improvising from experience — is the most important single procedural habit that the automotive mechanic must develop.

Done signal: You know the ATA chapter numbers for the 15 most commonly referenced systems in aircraft mechanical maintenance: Power Plant (71), Engine Fuel and Control (73), Engine Bleed Air (75), Engine Air (72), Engine Exhaust (78), Landing Gear (32), Wheels and Brakes (32 and related), Flight Controls (27), Hydraulic Power (29), Fuel (28), Pneumatics (36), Pressurisation and Air Conditioning (21), Oil System (79), Ignition (74), and Fire Protection (26). You understand the structure of an AMM task — the title and reference number, the references to related manuals (IPC for part numbers, CMM for component overhaul), the tools and consumables required, the safety precautions, and the numbered procedural steps. You understand what a cross-reference in the AMM task to the IPC means and how to use the IPC to find the correct part number for a replacement component.

3–4 hours

4. Understand Part-145 — the maintenance organisation approval and what it requires from every technician

Skill: The regulatory quality system that governs every maintenance action on a certificated aircraft

Why the field cares: Part-145 is the EASA regulation that approves organisations to carry out aircraft maintenance. Every airline engineering department, MRO facility, and component maintenance shop that works on EASA-registered aircraft holds a Part-145 approval. The approval defines what the organisation can do (its scope of approval), what procedures it must follow (documented in the Maintenance Organisation Exposition — the MOE), and what quality assurance systems it must maintain. For the technician, the most important Part-145 concept is personal regulatory accountability: every person who carries out aircraft maintenance under Part-145 is responsible for performing the work correctly, for documenting it accurately, and for not certifying work they have not personally performed or witnessed. The unlicensed technician who assists a licensed engineer must not sign off work they have not done. The licensed engineer who signs the CRS is personally accountable for that certification — and a CRS signed for work that was not performed in accordance with the approved data is a regulatory offence with licence consequences. The automotive mechanic who has worked in environments where shortcuts, informal modifications, and undocumented fixes are accepted practice must specifically develop the Part-145 quality discipline.

Done signal: You can explain the difference between the certifying staff (the Part-66 licensed engineer who signs the CRS) and the support staff (the unlicensed technician who performs the physical work under supervision). You can describe what the Maintenance Organisation Exposition is — the document that defines the organisation's procedures, its scope of approval, and its quality management system — and explain why every technician working in the organisation must follow the MOE procedures, not just the licensed engineers. You can describe what a Certificate of Release to Service is, who can sign it, and what the consequence is for the licensed engineer who signs a CRS for work not performed in accordance with the approved data.

2–3 hours

5. Understand human factors in aircraft maintenance — the Dirty Dozen, error preconditions, and the just culture reporting obligation

Skill: The safety culture and error prevention framework that every aviation maintenance employee must understand

Why the field cares: Human factors — the study of how individual, team, and organisational factors affect the likelihood of error in safety-critical tasks — is a mandatory Part-66 examination module (M9A) and a central element of the Part-145 quality system. The Dirty Dozen — the twelve most common preconditions for maintenance error (lack of communication, complacency, lack of knowledge, distraction, lack of teamwork, fatigue, lack of resources, pressure, lack of assertiveness, stress, lack of awareness, and norms) — provides the framework for recognising and managing the personal and organisational factors that lead to errors. The specific human factors scenario that most clearly distinguishes aviation maintenance from automotive is the just culture reporting obligation: when an aviation maintenance error is discovered — even one that did not result in an aircraft problem — the organisation's internal occurrence reporting system must be notified. In automotive, discovering that you installed the wrong part yesterday and correcting it quietly is standard practice. In aviation, the same discovery must be documented, reported, and investigated through the quality system — not as a punishment mechanism but as a safety learning mechanism. Every technician who enters aviation must internalise this reporting obligation before their first day on the job.

Done signal: You can name all twelve of the Dirty Dozen and describe what each means in a practical maintenance context, using examples from automotive or mechanical maintenance that you have personally experienced. You can describe a specific situation from your own experience in which one or more Dirty Dozen factors were present and explain what you did or what you should have done differently. You can explain what the just culture principle means for error reporting — that reporting errors in good faith does not result in punishment, but that deliberately concealing errors removes just culture protection. You understand why this reporting culture is specifically important in aviation in a way it is not in automotive.

3–4 hours

6. Understand aircraft engine oil systems — the specific differences from automotive and the aviation-grade oil requirements

Skill: The lubrication system that is the most directly familiar aircraft system for automotive and diesel mechanics — with specific differences

Why the field cares: Aircraft engine oil systems — for both piston engines and gas turbine engines — use specifically different lubricants from automotive applications, and the automotive mechanic who applies automotive oil knowledge to an aircraft engine will make safety-significant errors. Aircraft piston engines require ashless dispersant oil (grades such as W80, W100, W120, or W65/140 multigrade) — specifically not the automotive oils rated under the API classification system, which contain additives that can damage aircraft engine components, particularly copper-lead bearings. Gas turbine engines require synthetic lubricating oil (Type I or Type II synthetic turbine oil, meeting MIL-PRF-23699 or MIL-PRF-7808 specification) — again, completely different from any automotive or diesel lubricant. Using the wrong oil grade in an aircraft engine — a mistake that an experienced automotive mechanic might make from habit — is an airworthiness event requiring an oil drain, engine flush, and engineering investigation before the aircraft can fly. Beyond the oil grade, the oil system maintenance for aircraft differs from automotive in: the requirement to check oil consumption rate against the AMM specification (excessive oil consumption is a sign of internal engine problem requiring investigation, not a normal condition to be managed by topping up), the requirement to inspect the oil filter and screen for metal particles at every oil change (metal contamination is the primary indicator of internal engine distress), and the requirement to document the oil uplift quantity and the sample results in the aircraft technical log.

Done signal: You can explain why automotive engine oil (API-rated) cannot be used in a certified aircraft piston engine (the API additive package contains chemicals — specifically zinc dialkyldithiophosphate (ZDTP) — that are incompatible with the copper-lead bearings used in most aircraft piston engines, causing accelerated bearing corrosion). You can describe the difference between Type I and Type II gas turbine synthetic oil (Type II has higher viscosity and better high-temperature performance than Type I, and Type II is required for most modern turbofan engines). You can describe what the oil filter inspection for metal contamination looks for (ferrous and non-ferrous particles on a magnetic chip detector or in a filter element, with the specific AMM guidance for assessing whether the contamination level warrants further investigation or engine removal).

4–5 hours

7. Learn aircraft fasteners, hardware, and torque requirements — the specific differences from automotive

Skill: The hardware and assembly knowledge that is the most common source of errors for mechanics new to aviation

Why the field cares: Aircraft fasteners — bolts, nuts, washers, pins, and rivets — are specified more precisely than automotive hardware, and the failure to use the correct fastener in the correct location is an airworthiness event. The NAS and MS standardisation system (National Aerospace Standards and Military Standards) defines fasteners by material, thread form, head type, coating, and strength — and an NAS bolt of the same nominal dimensions as an automotive bolt may have completely different strength, different thread pitch, and different material. The torque value for an aircraft bolt is not a general automotive-style guideline — it is a specific value from the AMM or from the standard practices manual (typically Airbus AMM Chapter 20 or Boeing D6-54220), accounting for the bolt material, the installation location (whether the bolt passes through aluminium, steel, or composite), whether a lubricant or thread-locker is applied, and whether a self-locking nut or a castellated nut with cotter pin is used. Safety wiring — the stainless steel wire installed through the bolt heads of engine and flight-critical fasteners to prevent rotation — is a specific skill that aviation requires and automotive does not; the pattern, the wire diameter, the tension, and the pigtail length are all specified and are inspected during quality checks. The automotive mechanic who applies automotive torque intuition to aviation fasteners is likely to either under-torque or over-torque, both of which are maintenance errors with potential safety consequences.

Done signal: You can explain what a NAS 6604-20 bolt specification means — the NAS standard (NAS 6604 is a hex-head bolt in cadmium-plated alloy steel), the dash number indicating size and

4–5 hours

length (6 = 3/8 inch diameter, 04 = 4 grip length increment, 20 = 1.25 inch overall length in this series). You can describe the difference between a self-locking nut (all-metal or nylon insert type) and a castellated nut with cotter pin, and explain when each is appropriate (castellated nuts are used where the joint requires periodic disassembly, or where the all-metal locking nut cannot provide adequate locking torque). You can describe the basic safety wiring procedure — the direction of wire installation (such that the wire tends to tighten the bolt if pulled), the maximum span between anchor points, and the pigtail requirements (minimum 3 turns, cut to approximately 1/4 to 1/2 inch and bent back).

Time varies

8. Get practical exposure to aircraft maintenance — observation, shadowing, or volunteering before the first formal application

Skill: The environmental familiarity that makes job applications competitive and interviews credible
Why the field cares: The automotive mechanic who has never set foot airside, who has never seen an aircraft engine run, who has never held a Part-66 maintenance document, is applying blind to an environment they have never experienced. The most effective preparation — alongside the specific study tasks in this guide — is direct observation of aircraft maintenance in progress. Some MRO organisations and airlines offer engineering facility open days, work experience placements, or shadowing opportunities for people who are genuinely considering a career change. Flying clubs and general aviation maintenance organisations (CAA Part-M approved maintenance organisations for light aircraft) are specifically accessible for observation visits because they typically welcome genuine interest and are more informal in their operating environment than commercial MRO. The automotive mechanic who has arranged and completed one observation visit — who can describe what they saw in specific technical terms, what they found different from their automotive experience, and what specific questions the visit raised, is demonstrating the genuine commitment to the transition that every aerospace maintenance interviewer assesses.

Done signal: You have arranged and completed at least one opportunity to observe aircraft maintenance in progress — either at a general aviation maintenance organisation, through an MRO open day or shadowing programme, or through an aviation school or flying club that maintains its own aircraft. You can describe what you observed in technical terms (the type of work being done, the documentation being used, the tools in use, and the procedural discipline observed), what you found most different from your automotive experience, and one specific question or observation that the visit prompted about how your existing mechanical skills would or would not transfer.

1–2 hours

9. Understand the difference between aircraft maintenance and car maintenance — the procedural discipline that aviation requires

Skill: The attitudinal and procedural shift that is the most important cultural change in the transition
Why the field cares: The most experienced automotive mechanic must make a specific and deliberate attitudinal adjustment when entering aircraft maintenance — and the interview will specifically probe whether this adjustment has been made or is at least understood. The core difference: in automotive maintenance, professional judgement and experience are respected substitutes for following the documentation exactly. An experienced mechanic who has set the valve clearances on a hundred engines of the same type knows from experience what the clearance should be and can set it correctly without consulting the workshop manual. In aircraft maintenance, the same mechanic must consult the AMM, use the specified feeler gauges from the calibrated tool store, follow the specific sequence of adjustments specified in the procedure, and document the as-found and as-left measurements in the work order — even if they have set the same valves a hundred times before. Not because their experience is wrong, but because the regulatory framework requires that the work is traceable to an approved data reference, that the data reference specifies the procedure, and that the documented result demonstrates compliance. The mechanic who truly understands why this procedural discipline exists — who understands that the AMM is not insulting their competence but creating a legally auditable record — is ready for aircraft maintenance.

1–2 hours

Done signal: You can explain, in your own words and from genuine understanding rather than recitation, why aircraft maintenance requires documentation of every task and adherence to the approved data — specifically connecting the procedural requirement to the airworthiness consequence (the aircraft's legal airworthiness depends on compliance with its approved maintenance programme and data, and the CRS certifies that this compliance has occurred). You can describe a specific situation from your automotive career where you made a decision based on experience rather than the handbook, and explain what the equivalent decision in aircraft maintenance would require differently (consulting the AMM, finding the approved data, following the procedure step by step, and documenting the result). You can identify which aspect of the procedural discipline you find most challenging to adapt to and explain how you intend to address it.

2–3 hours

10. Learn the specific tools and measuring equipment used in aircraft maintenance — and the calibration requirements

Skill: The tool knowledge and calibration culture specific to aircraft maintenance

Why the field cares: Aircraft maintenance uses many of the same hand tools as automotive — spanners, sockets, ratchets, screwdrivers, pliers — but with specific additional requirements: all tools used on aircraft must be from a controlled tool inventory, signed in and out for each job, with a tool check completed before and after each job to ensure no tools have been left in the aircraft. The consequence of a foreign object left inside an aircraft — a spanner dropped into an engine intake, a screwdriver left inside a control surface — can be catastrophic, and the foreign object damage (FOD) prevention discipline is a fundamental part of aviation maintenance culture that has no automotive equivalent. Beyond the FOD discipline, calibrated measurement equipment — torque wrenches, pressure gauges, dimensional measuring tools — must be calibrated on a specified schedule and the calibration certificate available for inspection. A torque wrench that was last calibrated 18 months ago, when the calibration interval is 12 months, makes every torque application performed with it potentially invalid. The automotive mechanic who treats their torque wrench as a personal tool that they calibrate “when it seems off” must specifically develop the calibration management discipline that aviation requires.

Done signal: You can describe the Part-145 tool control procedure — the requirement to sign out tools at the start of a job, the tool check at completion of the job, and the reporting requirement if a tool cannot be accounted for. You can explain what FOD (Foreign Object Damage) is and why it is specifically important in aviation (a small object ingested into a running engine can cause catastrophic damage — the 2000 Concorde accident that killed 113 people was caused in part by a titanium strip on the runway that had fallen from an earlier aircraft, demonstrating that even objects not left by the current maintenance team can have catastrophic consequences). You can describe what a calibration certificate for a torque wrench specifies and why the calibration interval must be respected (if the calibration interval lapses, all torque applications since the last valid calibration are of unknown accuracy and may require re-torquing of flight-critical fasteners).

ROLE-SPECIFIC — TARGET AFTER FOUNDATIONAL TASKS

4–5 hours

11. For Cat B1 aircraft maintenance (the primary target): understand aircraft hydraulic systems — the specific differences from automotive

Skill: The aircraft hydraulic system that the B1 engineer maintains — specifically what differs from automotive

Why the field cares: The automotive and diesel mechanic who has worked on power steering systems, ABS hydraulics, clutch hydraulics, and air brakes understands hydraulic principles. Aircraft hydraulic systems add three specific differences that the mechanic must understand before their first aircraft hydraulic maintenance task. First, the operating pressure: commercial aircraft hydraulic systems operate at 3,000 psi (207 bar) for most existing aircraft and 5,000 psi (345 bar) for some newer aircraft — approximately 10–20 times the operating pressure of most automotive hydraulic systems.

4–5 hours

At these pressures, a pinhole leak in a hydraulic line produces a high-velocity fluid jet that can penetrate skin and cause serious injury. The specific safety procedures for working on aircraft hydraulics — the mandatory depressurisation procedures, the system isolation procedures, the PPE requirements — must be followed exactly. Second, the fluid: aircraft hydraulic systems use phosphate ester hydraulic fluid (Skydrol) rather than mineral oil, which is incompatible with the nitrile rubber seals used in automotive hydraulic systems (Skydrol requires EPDM or butyl rubber seals) and which requires specific handling precautions (Skydrol is irritating to eyes and skin). Third, the cleanliness requirement: aircraft hydraulic systems require much higher fluid cleanliness than automotive systems (NAS 1638 Class 6, compared to the ISO 4406 16–18 typical of automotive hydraulic systems) because the servo-valves and precision actuators in aircraft systems have much smaller clearances than automotive hydraulic components.

Done signal: You can describe the three key differences between automotive and aircraft hydraulic systems — operating pressure (3,000 psi vs. typically 150–200 psi in automotive), fluid type (Skydrol phosphate ester vs. mineral oil), and cleanliness requirement (NAS 1638 Class 6 vs. much less demanding automotive standard). You can describe the mandatory safety procedures before disconnecting a hydraulic fitting on an aircraft (the system must be depressurised through the specific procedure in the AMM, the hydraulic reservoir must be vented, and the depressurisation must be confirmed by gauging before any fitting is loosened). You can explain why using an automotive O-ring material in an aircraft hydraulic system seal application is an airworthiness error (the nitrile rubber will swell and deteriorate rapidly in contact with Skydrol, causing seal failure and hydraulic fluid loss).

3–4 hours

12. For GSE engineer roles: understand the specific electrical and mechanical systems of airport ground support equipment

Skill: The most immediately accessible aerospace role for automotive and diesel mechanics — and what it specifically requires

Why the field cares: Ground Support Equipment — the tractors, tugs, GPU units, belt loaders, passenger steps, catering vehicles, and de-icing rigs that service aircraft on the apron — is maintained by engineers whose primary skills are exactly those of the heavy vehicle and diesel mechanic. The GSE engineer's job is not aircraft maintenance in the Part-145 sense — the equipment being maintained is not the certificated aircraft itself, but the ground equipment that services it. This means: no Part-66 licence is required for most GSE maintenance roles; the documentation and regulatory framework is less demanding than aircraft maintenance (though not absent — IATA AHM 912 provides the ground handling manual for GSE maintenance standards); and the technical content is very close to the automotive and diesel mechanic's existing knowledge. The specific additions for GSE roles: the apron safety environment (airside driving licences, aircraft proximity procedures, fuel area restrictions, FOD prevention), the 400Hz electrical systems of ground power units (the specific three-phase 115V 400Hz AC power that aircraft use on the ground, generated by the GPU), and the increasingly important electrification of GSE (electric tugs, electric belt loaders, electric ground power units using lithium-ion battery systems).

Done signal: You understand what a Ground Power Unit (GPU) is — the equipment that supplies 115V 400Hz three-phase AC electrical power to the aircraft while its engines are shut down — and can explain why aircraft use 400Hz rather than 50/60Hz (smaller and lighter electrical components for equivalent power output). You know what an aircraft tug does and the specific safety requirement for tug operators (the tug operator must have an airside driving licence, must not exceed specified speeds on the apron, and must follow specific procedures for connecting to and disconnecting from the aircraft towbar). You have identified the specific IATA AHM 912 standard and understand that it provides the baseline quality requirements for GSE maintenance in the airline industry.

3–4 hours

13. For engine maintenance roles: understand the borescope inspection — what it finds, how it is performed, and what the findings mean

3–4 hours

Skill: The primary internal inspection tool for aircraft engines — with no automotive equivalent*

Why the field cares: The borescope inspection is the aircraft maintenance equivalent of the automotive compression test and oil pressure check — but it provides visual rather than performance evidence of engine internal condition. A borescope (a flexible or rigid optical probe with a camera) is inserted through access ports in the engine casing to view the internal surfaces of the combustor, the turbine stages, and the compressor stages without disassembling the engine. The borescope inspector looks for: combustor liner cracks or burn-through (which can result in hot gas ingestion into the engine structure if not addressed), turbine blade tip rubs (evidence of blade contact with the shroud, causing tip erosion and reducing engine efficiency), turbine blade cracking or oxidation damage (early indication of hot section distress), compressor blade erosion or FOD damage (causing loss of compressor efficiency and potential instability), and foreign object damage to any internal surface. The findings are assessed against the AMM serviceable limits — specific allowable crack lengths, tip rub dimensions, erosion areas — and the interpretation of a borescope finding against the AMM limits is a skilled judgement that aircraft engineers develop over many inspections.

Done signal: You can describe the purpose of a borescope inspection — to visually examine the internal surfaces of a gas turbine engine without disassembly, using flexible optical equipment inserted through access ports — and name four specific things a borescope inspector looks for (combustor liner cracks, turbine blade damage, compressor blade damage, and foreign object debris or damage). You can explain the difference between a serviceable finding (a crack or damage within the AMM allowable limits, documented and monitored) and an unserviceable finding (a crack or damage exceeding the AMM limits, requiring engine removal for repair). You understand why the borescope inspection is typically performed at specific intervals (at the engine on-wing service interval) and after specific events (bird strike, FOD ingestion, hard landing, or over-temperature exceedance).

4–5 hours

14. For mechanical systems maintenance: understand aircraft landing gear — the components, the maintenance tasks, and the safety requirements

Skill: The most mechanically complex aircraft system — and the one with the most direct automotive parallel

Why the field cares: The aircraft landing gear — the retractable undercarriage that the aircraft extends for takeoff and landing and retracts for flight — is the mechanical system with the most direct parallel to automotive suspension and braking systems. The main landing gear shock absorber (the oleo-pneumatic strut) uses a hydraulic oil and nitrogen gas combination to absorb landing energy — a concept that the hydraulic mechanic can understand from automotive pneumatic/hydraulic systems. The wheel and brake assembly uses hydraulic brake pistons and carbon or steel brake discs — directly analogous to automotive disc brakes, but with the addition of the anti-skid system (an electrohydraulic system that prevents brake lockup) and the specific aerospace specification brake materials. The tyre — a specific aviation tyre rated for the high speed and high load of aircraft operation, inflated to pressures of 150–200 psi (compared to 30–35 psi for automotive) — requires specific inspection, pressure monitoring, and inflation equipment. The specific safety requirements for landing gear maintenance: the aircraft must be supported on jacks before gear doors are opened or the gear is cycled hydraulically, and the specific jack points and jacking procedures from the AMM must be followed exactly.

Done signal: You can describe the main landing gear oleo-pneumatic strut — the hydraulic oil in the lower chamber that provides damping as the strut compresses, the nitrogen gas charge in the upper chamber that provides the spring force, and the metering pin or orifice that controls the damping rate. You can explain why the oleo strut extension (the amount the strut extends under the aircraft's weight) is checked and adjusted as part of routine maintenance — too little extension means insufficient energy absorption at landing; too much extension means the strut is over-inflated. You can describe the safety procedure for working on a landing gear system — specifically the requirement to put the aircraft on jacks (with the gear in the extended position) before any hydraulic landing gear operation, and the specific warning about never cycling the gear with maintenance personnel in the wheel well.

3–4 hours

15. Understand aircraft fuel systems — the specific differences from automotive fuel systems

Skill: The aircraft fuel system that contains, supplies, and manages the engine fuel — and the specific hazards involved

Why the field cares: Aircraft fuel systems differ from automotive fuel systems in three specific ways that the mechanic must understand. First, the fuel: commercial aircraft use Jet A or Jet A-1 kerosene rather than petrol or diesel. Jet A-1 has a higher flash point than automotive petrol (minimum 38°C, compared to approximately –43°C for petrol) but is similar in appearance and smell to diesel and can be confused with it. Using the wrong fuel type — a fuel contamination event — can cause engine damage and is a serious airworthiness and safety event requiring specific procedures. Second, the fuel tank location: aircraft fuel is stored in the wings and sometimes the fuselage or tail in integral tanks (the wing structure itself forms the tank boundary) — not in a separate tank that can be unbolted and replaced. Working inside a fuel tank requires specific confined space entry procedures, specific breathing apparatus, and specific bonding and earthing to prevent electrostatic ignition of fuel vapour. Third, the fuel management system: commercial aircraft manage fuel between multiple tanks using fuel transfer pumps, crossfeed valves, and fuel quantity indication systems — a complexity that has no automotive equivalent and that requires specific AMM guidance for every fuel system task.

Done signal: You can explain why fuel tank entry is classified as a permit-required confined space entry in aviation maintenance (the fuel vapour in a partially emptied tank can reach explosive concentrations, and the confined geometry can trap the vapour without effective natural ventilation). You can describe the bonding and earthing requirements before opening any aircraft fuel system component (the aircraft must be bonded to an earthing point before any fuel cap or fitting is opened, to prevent electrostatic sparking from dissimilar-material contact). You can describe the fuel contamination event procedure — what must happen if a wrong fuel type is discovered in an aircraft tank (the flight must be immediately stopped if not yet commenced, all fuel must be drained, the tanks must be flushed, and an engineering investigation and report must be completed before the aircraft can return to service).

3–4 hours

16. Understand aircraft composite materials — how to inspect and handle them without damaging them

Skill: The dominant structural material of new aircraft — and the one most different from automotive metallic repair

Why the field cares: The Airbus A350 is 53% CFRP (carbon fibre reinforced polymer) by weight; the Boeing B787 is approximately 50% composite. Every commercial aircraft manufactured today has substantial composite content, and the mechanical technician who cannot safely work on composite structures — who drills at the wrong speed and delaminates the material, or who applies the wrong solvent during cleaning and degrades the matrix resin — is creating airworthiness hazards. The specific composite knowledge that the mechanical technician must develop: composites fail differently from metals (an aluminium panel that is dented is visibly damaged; a CFRP panel struck by the same impact may have no visible external damage but severe internal delamination — the SRM process for assessing composite damage always requires NDT, not just visual inspection); composites are sensitive to specific chemicals (many paint strippers, hydraulic fluids, and cleaning solvents attack composite matrix resins, and the AMM specifies exactly which cleaning agents and solvents are approved for use near composite structure); and composite repairs require specific approved materials and cure procedures that differ from metal repairs.

Done signal: You can explain why a visual inspection alone is insufficient for assessing impact damage to a CFRP composite panel (the resin matrix can delaminate internally without any visible surface evidence — the internal delamination is detectable only by ultrasonic C-scan or tap testing). You can describe the tap test — the qualitative composite inspection technique of tapping the surface with a coin or purpose-made tap hammer and listening for the change in acoustic response that indicates a delamination — and explain its limitations (it is sensitive only to near-surface delaminations within approximately 6mm of the tapped surface, and it cannot provide a reliable size estimate for the delamination). You understand why using an unapproved solvent near a composite panel for cleaning

3–4 hours

is an airworthiness issue (if the solvent penetrates the surface through a porosity or a surface defect, it can attack the resin matrix and degrade the structural properties of the panel).

2–3 hours

17. Understand the aircraft technical log and maintenance record requirements

Skill: The documentation that makes every maintenance action legally valid

Why the field cares: The aircraft technical log — the document that travels with the aircraft on every flight and records every defect, every maintenance action, every fuel and oil uplift, and every Certificate of Release to Service — is the legal health record of the aircraft. Every entry in it is permanent, it forms the basis for the airworthiness authority's oversight of the aircraft, and it is a primary document in any accident or incident investigation. The mechanical technician must understand what a technical log entry must contain (the date, the description of the work, the approved data reference, the parts used with their part numbers and batch numbers, the certifying engineer's name and licence number, and the CRS statement), what it must not contain (white-out correction fluid — errors must be corrected by a single line through the incorrect entry, with the correction written alongside and initialled), and what the regulatory consequence is of a falsified technical log entry (falsifying an aircraft technical log is a criminal offence under aviation law, not merely a regulatory infringement).

Done signal: You can state the six required elements of a maintenance record entry in the aircraft technical log (date, description of work, approved data reference, parts used with part numbers, certifying staff's name and Part-66 licence number, and CRS statement). You can describe the correct procedure for correcting an error in a technical log entry (single line through the error, correction written alongside, initialled by the certifying staff member who made the original entry — obliterating the original entry is specifically prohibited). You understand why falsifying a maintenance record is a criminal matter rather than merely a regulatory one — the aircraft technical log is a legal document that forms the basis for airworthiness certification, and a false entry misleads the certifying authority and anyone who relies on it.

2–3 hours

18. Build approximate knowledge of key aircraft maintenance parameters — torques, pressures, temperatures, and clearances

Skill: The approximate engineering awareness that helps the experienced mechanic recognise when something is wrong before consulting the manual

Why the field cares: The experienced aircraft maintenance engineer has developed an approximate awareness of typical values — the kind of knowledge that allows a preliminary sanity check before the AMM value is looked up. A landing gear tyre inflated to 20 psi is obviously wrong (the typical value is 150–200 psi); an engine oil level above the maximum mark on the dipstick is obviously wrong; a torque wrench set to 5 ft-lb for a large structural bolt is obviously wrong. This approximate awareness — which the automotive mechanic has already developed for automotive systems — must be specifically rebuilt for aircraft systems, because the typical values are different. The correct approach in aviation is always to verify the specific AMM value before applying any torque, pressure, or measurement — but the approximate awareness allows the experienced engineer to catch obvious errors before they become maintenance events.

Done signal: You can state, with the explicit caveat that these are approximate guidance values to be verified in the AMM for the specific aircraft and task, approximate typical ranges for: landing gear tyre pressure (main gear: 150–220 psi; nose gear: 120–180 psi), landing gear strut nitrogen pre-charge pressure (approximately 1,500–1,800 psi for most commercial aircraft, checked with aircraft on jacks), aircraft hydraulic system operating pressure (approximately 3,000 psi for most commercial aircraft), engine oil quantity check requirement (checked within 30 minutes of engine shutdown and before any oil is added, to allow the oil to drain back from the engine internal galleries), and structural bolt torque range for a 3/8 inch diameter NAS bolt in aluminium (approximately 140–200 inch-pounds in standard conditions — very different from the equivalent automotive torque specification). You state clearly that every value must be verified in the AMM or relevant aircraft documentation before any maintenance action.

POSITIONING AND PRESENTATION

19. Prepare a specific answer to "Why aircraft maintenance — and what from your automotive/diesel background makes you ready?" 2–3 hours

Skill: The interview answer that turns your trade background into an immediately credible aviation maintenance foundation

Why the field cares: The aviation maintenance interviewer who sees an automotive or diesel mechanic application has a specific question: does this person understand what is the same between their current work and aviation maintenance, what is specifically different, and do they have the right attitude toward the differences? The answer "I have spent eight years as a diesel technician diagnosing faults on heavy vehicles, including complex electrical faults, hydraulic system faults, and turbocharged engine problems. I understand that aircraft maintenance is a different environment — specifically the regulatory framework of Part-145, the requirement to work to approved data without improvisation, the documentation discipline, and the Skydrol hydraulic fluid that requires different seals from what I've used. I've been studying for the Part-66 B1 examinations and have passed Modules 3 and 6. I arranged a visit to [MRO] last month to observe their base maintenance operation, and what I found most valuable was seeing how the AMM fault isolation logic worked compared to the systematic diagnostic approach I use — the structure is the same, the documentation requirement is different. My turbocharged diesel experience is my strongest preparation for gas turbine theory — I understand how boost pressure and exhaust gas temperature are related, and the Part-66 M11A gas turbine module has been the most intuitive content for me to study" is demonstrating the honest, specific, prepared transition that aviation maintenance managers respond to.

Done signal: You have a 90-second answer that: names the specific automotive or diesel experience most relevant to the target aerospace role; explicitly acknowledges the three most important specific differences between your current work and aviation maintenance (approved data adherence, documentation discipline, and the specific regulatory framework); demonstrates that you have taken at least one concrete preparatory action (studying for Part-66, visiting an MRO, or reading about the specific aircraft type); and explains what aspect of your automotive or diesel experience specifically excites you about the aerospace application (turbocharged diesel experience for gas turbine, hydraulic fault diagnosis for aircraft hydraulic systems, or heavy component removal for power-plant maintenance).

20. Map your specific trade background to the right first aerospace role and employer — the entry route that fits your skills and timeline 4–6 hours

Skill: The targeting decision that determines whether the transition happens in 6 months or 3 years

Why the field cares: The automotive and diesel mechanic has multiple entry routes to aerospace, and the right one depends on their specific background, their geographic location, and their tolerance for the financial cost of full-time training. The four primary routes are: direct entry to a Part-147 approved training organisation that runs a B1 conversion programme (full-time, typically 12–18 months, expensive but fastest to licence); direct hire as a Cat A or Cat B1 unlicensed technician at an MRO or airline that supports Part-66 study in parallel with employment (longer to licence — 3+ years — but income during the process); GSE engineer at an airport ground handler or airline (aviation environment, relevant experience, no Part-66 required immediately, income during Part-66 study); or a direct-entry aerospace apprenticeship that specifically targets engineering technicians from automotive or engineering backgrounds (most airlines that run engineering apprenticeships will accept applicants with trade backgrounds — British Airways Engineering, Rolls-Royce, BAE Systems).

Done signal: You have identified the specific entry route that is most viable for your circumstances — considering your geographic location (is there a Part-147 training organisation nearby? is there an MRO or airline engineering base within commuting distance?), your financial situation (can you sustain a period of full-time training, or do you need to earn while studying?), and your current qualifications (do you hold recognised mechanical qualifications that a Part-147 programme would credit?). You have contacted at least one specific employer or training provider to discuss entry requirements and have a realistic timeline for completing the Part-66 examinations and the required

experience hours. You have at least one application in active progress — whether to a training programme, a GSE role, or an unlicensed technician position.

4–6 hours

SUMMARY REFERENCE TABLE

#	Task Description	Target Priority
1	Part-66 B1.1 pathway — modules, module mapping to automotive knowledge, timeline	All / Do first
2	Gas turbine engine — Brayton cycle, four sections, differences from diesel	All
3	ATA chapter system and AMM structure — navigation, task format, IPC	All
4	Part-145 framework — MOE, CRS, personal accountability, the certifying staff role	All
5	Human factors — Dirty Dozen, just culture reporting, personal story	All
6	Aircraft engine oil — ashless dispersant, synthetic turbine oil, differences from automotive	All
7	Aircraft fasteners and hardware — NAS/MS standards, torque, safety wiring	All
8	Practical aviation observation — MRO visit, GA maintenance observation	All / Critical
9	Procedural discipline shift — why approved data compliance is non-negotiable in aviation	All
10	Aircraft tool control and calibration — FOD discipline, calibration intervals	All / Critical
11	Aircraft hydraulic systems — 3,000 psi, Skydrol fluid, cleanliness requirements	B1 target / All
12	GSE engineer role — GPU 400Hz, apron safety, eGSE, IATA AHM standards	GSE entry route
13	Borescope inspection — what it finds, how it is performed, AMM assessment	Engine maintenance roles
14	Aircraft landing gear — oleo strut, wheel and brake, jacking safety	Mechanical maintenance
15	Aircraft fuel systems — Jet A-1, tank entry safety, contamination procedures	All
16	Aircraft composite materials — impact damage, tap test, chemical compatibility	All
17	Aircraft technical log — required elements, error correction, regulatory significance	All
18	Approximate aircraft maintenance values — tyres, hydraulics, oils, torques	All
19	“Why aviation — what from my automotive background?” — 90-second positioned answer	All
20	Entry route selection and first application — training vs. employment vs. GSE pathway	All

THE STRUCTURAL REALITY OF THE AUTOMOTIVE AND DIESEL MECHANIC-TO-AEROSPACE TRANSITION

The automotive and diesel mechanic who enters aviation maintenance is not entering an unfamiliar world. They are entering a more demanding version of the world they already inhabit — a world where the mechanic's judgement, their systematic diagnostic approach, and their mechanical competence determine the outcome of their work. The aircraft is not a car, but the person who maintains it has the same fundamental responsibility as the person who maintains any machine on which people depend: to understand the system, find the fault, repair it correctly, verify the repair, and leave the system in a condition that is safe for its next operation.

What is genuinely different is the consequence of getting it wrong, the regulatory framework that exists specifically because of that consequence, and the professional credentialing system that gates access to the most consequential act in aviation maintenance — the signature on the Certificate of Release to Service. The CRS is the automotive mechanic's equivalent of handing the keys back to the customer, but with one difference: the aircraft that the CRS authorises to fly may carry 200 people to 35,000 feet, and the certifying engineer's name is permanently attached to that authorisation. That is a different level of professional accountability than any automotive workshop provides, and it is exactly why the Part-66 licence requires 3 years of supervised experience and formal examination, not just a competency test.

The automotive mechanic who understands this — who sees the procedural discipline of aviation not as bureaucratic overhead but as the engineering response to a different level of consequence — is already making the most important mental adaptation that the transition requires. The mechanic who has spent years doing their automotive work with genuine care — who always torqued to specification, always documented what they did, always tested after repair, and always reported problems they found regardless of whether they caused them — is closer to aviation-ready than they know. The examinations are learnable. The gas turbine theory builds on thermodynamic intuition that diesel mechanics already have. The AMM navigation is a skill that develops within months of regular use. The Skydrol fluid compatibility and the NAS fastener system are specific and finite additions to a mechanical knowledge base that is already strong.

The transition is not easy — the Part-66 examination programme requires genuine study, the experience requirement takes time, and the procedural discipline requires deliberate attention. But it is specifically achievable for the automotive and diesel mechanic, more achievable than for almost any other background except the electrical technician, because the foundational mechanical competence is already in place. The aircraft maintenance industry knows this. That is why it runs apprenticeship programmes that specifically target trade-qualified mechanics, why its employers consistently list “mechanical trade background” as a valued entry qualification for Cat A and support technician roles, and why its licenced engineers, when asked where their best new recruits come from, regularly mention the motor trade alongside the aviation apprenticeship programmes. The aircraft industry needs people who know what it feels like to have oil on your hands, to trace a fault systematically through a complex system, and to hand back a machine that works correctly when it didn't before. That is exactly what the automotive and diesel mechanic knows. The aerospace sector is the most demanding and most consequential context in which to apply that knowledge — and for the mechanic who is ready to make that application seriously and with the specific preparation this guide requires, it is a career transition that is both viable and worthwhile.

BREAK INTO AEROSPACE